

Variance proof

Group 4 SDS 188

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Show that:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - \frac{1}{n} \left[\sum_{i=1}^n x_i \right]^2 \right)$$

Proof:

$$\begin{aligned} s^2 &= \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \sum_{i=1}^n \left(x_i^2 - 2\bar{x}x_i + \bar{x}^2 \right) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + \sum_{i=1}^n \bar{x}^2 \right) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - 2\bar{x}(n\bar{x}) + n\bar{x}^2 \right) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \right) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - n \left[\frac{1}{n} \sum_{i=1}^n x_i \right]^2 \right) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - \frac{n}{n^2} \left[\sum_{i=1}^n x_i \right]^2 \right) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - \frac{1}{n} \left[\sum_{i=1}^n x_i \right]^2 \right) \end{aligned}$$